

Karsten Sperling Opdal

Developing Embedded Systems into the future of AI

Contact details

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Summary

An embedded Linux specialist bringing embedded platforms to life — reliably and securely.

30+ years spanning Nokia handsets, Infineon baseband platforms, and medical simulation devices. He owns the full stack from silicon to screen: Yocto build systems, kernel bring-up, real-time C++ services, Qt/QML interfaces, CI/CD pipelines, secure OTA deployment, and security hardening (secure boot, RED, FIPS 140). Builds his own tools — from AI-powered crash analyzers to HSM-backed certificate portals. AI is deeply integrated into his engineering workflow.

Ready to help design and architect the next embedded system, where security is the foundation.

Copenhagen, Denmark | Open to remote across Europe

Core Skills

C · C++ (C++23) · Embedded Linux · Yocto · Linux kernel · Qt/QML (Qt 3–6) · Python · Rust · WiFi/Bluetooth · CI/CD · GitHub Actions · Boost · Azure · Secure boot · FastAPI · Tauri · AI-assisted engineering

Languages

Danish: Native speaker (written and spoken)

English: Fluent — second language used throughout career, full professional proficiency

Norwegian: Fluent understanding (spoken and written), conversational speaking — 25 years' exposure, Norwegian spouse

Swedish: Strong understanding (spoken and written), near-conversational speaking — 30 years' exposure, neighbouring country

German: Good understanding (spoken and written), speaking somewhat rusty — 10 years during education

Experience

Laerdal Medical

Senior Software Engineer Consulting

October 2010 – February 2026 (15 years 5 months)

- Senior Embedded Software Engineer / Build System Architect for SimPad devices / simulators and SimMan3G product lines.
- Lead maintainer of Yocto/OpenEmbedded build platform delivering production Linux across ARM32, ARM64, x86-32, and x86-64. Evolved through six major Yocto releases (Rocko to Scarthgap). Migrated from Ubuntu to custom distribution with RAUC A/B OTA updates and Azure Key Vault HSM signing.
- Architected multi-platform CI/CD pipelines with GitHub Actions, SBOM generation, and automated CVE tracking for medical device compliance.
- Linux kernel configuration, debugging, patching, and Device Tree customization across i.MX6, i.MX8M Plus, and Intel x86 — board bring-up, peripheral enablement, pin muxing, clock trees, and driver troubleshooting.
- Developed real-time simulation engine controlling medical manikins (SimMom, SimBaby, MammaAnne, SimMan ALS) via CAN bus or IOC controller and physiological modelling. C++ (up to C++23) with Boost.
- Designed and developed SimPad touch-screen UI using Qt 4–5 with QML for embedded interfaces.
- Proficient in C#, used for LLEAP development.
- WiFi connectivity and ALSA audio pipelines for clinical simulation. Bluetooth expert with patches contributed to BlueZ and Qt Bluetooth stacks.
- Applied EU Radio Equipment Directive (RED) compliance — hardened Linux kernel configuration, implemented secure boot, and built a CVE checker tool that analyzed whether kernel config options introduced known vulnerabilities or whether existing patches already addressed them.
- Created SimServer Imager (Qt6/C++/QML, based on Raspberry Pi Imager) for managed WIC/VS1/SPU deployment and CDN distribution.
- Built VEX Kernel Checker — AI-assisted Kernel CVE analysis integrated with Dependency-Track.
- Built SSH Certificate Portal (FastAPI/Python) — self-service time-limited SSH certificates with Azure AD OIDC, HSM-protected CA signing (RSA/ED25519/ECDSA), deployed on Azure App Service.
- Designed Azure Key Vault HSM signing workflow for RAUC bundles and secure boot — non-exportable keys, automated renewal via Azure Automation, multi-environment CI integration.
- Built Jira Analyse Companion — AI-powered diagnostic platform using Claude/Gemini/OpenAI for crash analysis across SimPad, LinkBox, and CAN firmware. Jira automation, source context integration. Delivered as CLI, VS Code extension, and Tauri desktop app.

Skills: C++ · C++23 · Boost · C# · Rust · Python · Bash · JavaScript · TypeScript · Yocto · Linux kernel · Device Tree (DTB) · ARM (i.MX6/i.MX8) · x86 · Qt 4 · Qt 5 · Qt 6 · QML · FastAPI · RAUC · Azure · GitHub Actions · CAN bus · ALSA · WiFi · Bluetooth · BlueZ · Tauri · Secure boot · RED · Security · GDPR · FIPS 140 · NIST SP 800-171 · Unit testing · TDD · OOD · System architecture · Software architecture · WCF · XAML · MAUI

Oscilloscope

Senior Software Engineer Consulting

October 2010 – June 2019 (8 years 9 months)

- Freelance

Nokia

Software Engineering Specialist

April 2010 – October 2010 (7 months)

- I excelled in identifying and resolving complex firmware issues at Nokia, significantly enhancing product performance.
- Specialized in debugging intricate hardware/software interaction bugs on mobile platforms.
- Successfully fixed a battery drain issue that had persisted for years, improving battery lifetime by 30%.
- Developed critical skills in embedded systems analysis and problem-solving within a leading technology company.

Skills: C · C++ · Linux · BSD · Unix · Embedded firmware · Debugging · Lithium battery management · Teaching · Developer support

Nokia Mobile Phones

Senior Software Engineering Consultant

September 2008 – February 2010 (1 year 6 months)

- Provided expert consultation on the S40 platform software, focusing on bug fixing and error correction.
- Debugged firmware issues across the S40 software stack to enhance performance and reliability.
- Collaborated with cross-functional teams to ensure timely resolution of software defects, improving user experience.

Skills: C · C++ · S40 platform · Firmware debugging · Visual Studio plugin · BSD · Linux · Unix · Bash

Infineon

Software Expert

October 2007 – September 2008 (1 year)

- Developed the Nokia 2110 low-cost phone platform during an expat assignment in Copenhagen.
- Optimized embedded software for resource-constrained hardware, ensuring efficient use of memory.
- Delivered high-quality software under tight deadlines, focusing on minimizing resource usage.

Skills: C · Embedded software · Memory optimization · Linux · Unix · BSD · Expat · Mentoring

Thorsø Data

Software Expert

October 2007 – September 2008 (1 year)

- In house Senior Contract Software Engineer

Tang-Data A/S

Senior Software Engineer

March 2007 – October 2007 (8 months)

- Developed a comprehensive veterinary CRM system utilizing Qt 3 and Qt 4, enhancing client management.
- Provided on-site support for veterinary computer setups, ensuring seamless CRM hardware installations.
- Collaborated with cross-functional teams to address customer needs and improve system functionality.

Skills: C++ · Qt 3 · Qt 4 · Linux · CRM · OOD · Teaching end users

Nokia

Senior Software Engineer

September 2000 – February 2007 (6 years 6 months)

- Enhanced the software development lifecycle at Nokia through innovative UI development.
- Developed internal tools that streamlined handset software testing processes.
- Enabled mobile phone development in a simulated environment using Microsoft Visual Studio.
- Developed a music player database using SQLite on a memory-constrained S30 embedded device, using test-driven development as the player and UI were not yet available.

Skills: C · C++ · Visual Studio · wxWidgets · Qt 3 · Unix · BSD · Simulation · UI development · SQLite · TDD · OOD · WCF · XAML

EC-Soft Danmark A/S

Senior System Software Engineer

October 2000 – December 2001 (1 year 3 months)

- Consultant work for Nokia Denmark A/S developing S30 phones

Telenor

Contract Software Engineer

March 2000 – August 2000 (6 months)

- Developed a web service enabling mobile access to Outlook email and calendar via WAP, one of the earliest mobile web integrations.

Skills: Microsoft Visual Basic# (.NET) · REST API · Web Server · WAP

EC-Soft Norge AS

Senior Software Engineer

March 2000 – September 2000 (7 months)

- In house Software Consultant

Nokia

Contract Software Engineer

May 1999 – February 2000 (10 months)

- Developed user interfaces for Nokia's iconic handset models, including the 3210 and 3310.
- SMS domain expert with full knowledge of 2G protocols. Collaborated with a fellow developer to implement concatenated SMS messaging.
- Integrated T9 input support, improving text input efficiency for users.
- Improved memory management on the S30 platform, optimizing resource usage on constrained hardware.

Skills: C · Unix · BSD · 2G protocols · SMS expert · T9 · UI development · S30 platform · Memory optimization

EC-Soft Danmark A/S

Software Consultant

May 1999 – February 2000 (10 months)

- In house software consultant

Vizion Factory ApS

Software Engineer

December 1997 – April 1999 (1 year 5 months)

- Played a key role in software development and early web solutions, contributing to the burgeoning internet landscape.
- Built applications that addressed emerging needs and developed a training engine paired with a web interface to streamline user interaction.
- Developed a web browser in Delphi Pascal to render HTML-based training materials.
- Created screensavers using DirectX for 3D rendering.

Skills: Delphi Pascal · OOD · Windows · Windows COM+ · DirectX · Assembler x86 · Web development

Greve Kommune

Network System Specialist

January 1993 – October 1997 (4 years 10 months)

- Set up new networks to enhance connectivity and efficiency within the organization.
- Led network evolution:
 - 1 Mbit coaxial LAN
 - 10 Mbit (10BASE-T)
 - 100 Mbit UTP (100BASE-TX)
- Provided training to employees on computer use, fostering a tech-savvy workplace.
- Maintained hardware by adding and replacing components, ensuring optimal performance.
- Built a database system for time registration and contractor management using Microsoft Visual Access.
- Created celebration cards and information posters using CorelDRAW.
- Developed WordPerfect macros to standardize letter templates for Greve Kommune's new visual identity.
- Participated in transition from WordPerfect to Microsoft Office (Word, Excel), including user training and template migration.

Skills: Ethernet Protocol · Coaxial LAN (1 Mbit) · 10BASE-T · 100BASE-TX UTP · TCP/IP · Managed networking · Hardware · WordPerfect · Microsoft Office · Microsoft Visual Access · Database development · CorelDRAW · Teaching

Military Service

Recruit

February 1992 – October 1992 (9 months)

- Danish Army, HAWK-eskadrille at Stevnnsfortet, Zealand

Opdal Enterprise

Owner

September 2008 – Present (17 years 7 months)

- Køge Municipality

Home development

Developer

April 1983 – Present (43 years)

- Developed Game for C-64, demos for C-64 and Amiga and a task / thread / resource manager for Amiga OS.
- Developed music editor software C-64, Amiga and PC Dos.
- Currently developing Android application for learning purpose and application with Rust and Svelte

Skills: C-64 · Amiga · Assembler 6502/68000 · C++ · C# · Python · Rust · Svelte · Android · GitHub · Azure · AWS · Web API · OOD

Hjemmeværnet

Volunteer

April 2002 – Present (24 years)

- Danish Home Guard, Hjemmeværnet, Køge, Zealand
- Trained in basic military skills and regularly participate in exercises and operations to support national defense and civil authorities.

Github repositories

<https://github.com/palettéguy/profile>

<https://github.com/Laerdal/vex-kernel-checker>

<https://github.com/Laerdal/linux-fslc>

<https://github.com/Laerdal/meta-dependencytrack>

<https://github.com/Laerdal-Medical/simserver-imager>

Education

Software development

Niels Brock

Bachelor's Degree, Computer Science

August 1990 – January 1992

Niels Brock

Bachelor's Degree, Business/Commerce, General

August 1988 – June 1990

EFG Handel og Kontor

Basic business school

August 1987 – Juni 1988

Krogaardskolen

High school diploma

1976 – 1987